

Abundance and dimensions of naturally occurring macropores along stream channels and the effects of artificially constructed large macropores on transient storage

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Abstract: Macropores are connected void spaces in the subsurface and can act as preferential flow paths for groundwater transport, but their dimensions and distribution patterns have not been well characterized in stream banks and hyporheic sediments. We ran field surveys in 5 streams of varying size, bed slope, and watershed land use in the Appalachian Mountains of southwestern Virginia. Natural surface-connected macropores were nearly ubiquitous. Macropores in this region arise by a variety of mechanisms, including soil piping, tree root decay, erosion around hardened structures, macroinvertebrate burrows, and invertebrate burrows. Macropore openings were slightly wider than tall, with median cross-sectional widths and heights of 3.5 cm and 3.0 cm, respectively. The median and maximum macropore lengths to first bend were 15.0 cm and 120.5 cm into the bank, respectively. True (tortuous) macropore lengths probably are often greater, and methods to map macropores are needed. Median interspacing of macropores across all streams was 0.38 m, but mean interspacing was 1.12 m, indicating that macropores were clustered in space. Macropores were inundated at different times because of differences in their heights on the stream bank, channel geometry, and stream stage. HEC-RAS modeling of channel hydraulics indicated that only 1 to 32% of macropores were inundated by channel water at base flow, whereas up to 97% were inundated during the largest storms. We ran conservative-tracer injection experiments at base flow in a 30-m reach of a small tributary to 1 study stream. We calculated transient-storage parameters for 2 treatments: 1 without macropores and 1 with artificially created macropores. Our results suggest that constructed macropores weakly increased transient storage (channel:storage zone cross-sectional area [A_s/A] by 15.0% and fraction of median travel time spent in the transient-storage zone [F_{med}] by 31.0%) and may affect surface-water–groundwater interactions. Many macropore openings were above typical baseflow water levels, so macropores may enhance bank storage, a possibility that bears further research.

Key words: connected void spaces, hydraulic conductivity, field survey, hydroperiods, hyporheic exchange, transient storage

The hyporheic zone is the region where surface water and groundwater interact beneath and adjacent to stream channels (Triska et al. 1989). Hyporheic flow paths can extend vertically (e.g., weir-induced hyporheic exchange), laterally (e.g., intrameander flow), or longitudinally (e.g., along paleochannels, through mid-channel bars) along the stream channel depending on stream morphology and hydrologic conditions (Poole et al. 2006, Burkholder et al. 2008, Hester and Doyle 2008, Revelli et al. 2008). Surface water generally is high in dissolved O₂ and nutrients, whereas ground water is generally low in dissolved O₂, high in inorganic solutes,

and has low velocities relative to surface water (Triska et al. 1989, Brunke and Gonser 1997). The interaction of these 2 water bodies creates unique conditions that can provide various functions for the surface stream including temperature regulation, nutrient cycling, pollutant buffering, and habitat creation (Brunke and Gonser 1997). Hydraulic conductivity (K) is a key variable controlling hyporheic exchange rates (Findlay 1995, Valett et al. 1996, Hester and Doyle 2008, Boulton et al. 2010, Hester and Cranmer 2014, Menichino and Hester 2014). Regions of relatively high K act as preferential flow paths for groundwater flow and can

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dominate water and solute transport through porous media near streams (Fox et al. 2009, 2011) and control bank storage (Kondolf et al. 1987).

Macropores are a type of preferential flow path devoid of sediment and larger than the local pore space size of the media. Macropores including those formed from the dry fracture of soils (Blake et al. 1973), earthworm burrows (Edwards and Loft 1977, Edwards et al. 1979), and the decay of plant roots (Aubertin 1971, Beasley 1976) can dominate subsurface transport. Macropores have not been studied extensively in the hyporheic zone, but a growing number of studies suggest they are common (Bohlke et al. 2007, Gormally et al. 2011, Newman and Keim 2013). For instance, macropores are created by burrowing fauna, including macroinvertebrates (e.g., stonefly nymphs, mayflies, worms, crayfish, crabs, mussels, clams), birds (e.g., belted kingfisher, bank swallows), mammals (e.g., mink, beaver, coypu, muskrat), and reptiles (e.g., box turtles) (Shields and Kelly 1997, Allen and Vaughn 2009, Xin et al. 2009, Brown and Brown 2011, Wright et al. 2011), and bioturbation by benthic macroinvertebrates (Mermillod-Blondin et al. 2004, Mermillod-Blondin and Rosenberg 2006, Nogaro et al. 2006, DiStefano et al. 2009, Song et al. 2010). However, no systematic evaluation has been made of all macropore formation mechanisms along example streams or rivers. Thus, the density and distribution of macropores are generally unknown. Furthermore, no investigator has quantified how frequently such macropores are inundated, so their temporal hydrologic characteristics also are poorly understood.

The spatial characteristics (e.g., frequency, size, interconnectivity) of macropores are important for estimating flow rates through macropores (Matthai and Belayneh 2004). Spatial characteristics may vary widely with macropore creation mechanism and local stream conditions (e.g., hydrology, geology). For example, crayfish burrows can start in the channel, extend well into the banks in the groundwater zone and through the parafluvial zone (Williams et al. 1974). Soil pipes are a type of macropore formed by subsurface erosion and can connect the floodplain and hill slopes to the surface channel (Jones 1971, Holden 2005, Jones and Cottrell 2007, Fox and Wilson 2010). Noguchi et al. (1999) found that most macropores in the unsaturated zone of a forested hill slope were 2 to 62 cm in length and discontinuous. These studies provide spatial dimensions for certain types of macropores, but the authors did not systematically quantify the characteristics and spatial distribution of all types of macropores directly connected to the surface stream channel (surface-connected) along a range of streams. Thus, how widespread macropores are, even within an individual physiographic province, is not known.

Macropores (in particular, soil pipes) can facilitate flow from hill slopes toward streams, providing a significant

portion of flow to streams during storms (Jones and Crane 1984). Surface-connected macropores within a meander bend can substantially increase hyporheic flow rates and exchange with surface water (Menichino et al. 2014). These macropores may increase solute transport between the stream and riparian zone (Angier and McCarty 2008, Gormally et al. 2011, Newman and Keim 2013). However, these studies focused on sediment-scale hydraulics rather than larger-scale hydrologic effects, such as baseflow transient storage. At base flow, applying the transient-storage method may effectively quantify the effect of surface-connected macropores on solutes moving downstream in surface water. The transient-storage method measures the effect of subsequent step increases and decreases of dissolved conservative tracer injected into a stream at downstream locations, thereby quantifying exchange with both surface (dead zone) and subsurface (hyporheic exchange) storage zones (Bencala and Walters 1983, Runkel 2002). The method has been used to quantify transient storage resulting from in-stream baffles, streambed topography, and stream restoration (Ensign and Doyle 2005, Bukaveckas 2007, Gooseff et al. 2007). Macropores probably existed in streams where transient-storage experiments have been run in the past, but to our knowledge, the effect of macropores on transient storage has not been quantified.

We conducted a field study of the spatial characteristics of natural surface-connected macropores and tested the effect of artificially constructed macropores on transient storage. Our 1st objective was to survey and quantify reach-scale abundance and dimensions (diameter, depth, elevation within the channel) of natural macropores across 5 streams that represent a range of stream order, land use, and geomorphic conditions typical of the Valley and Ridge physiographic province in the Appalachian Mountain region of southwestern Virginia. Our 2nd objective was to determine temporal variation of natural macropore inundation (i.e., macropore hydroperiods) at the reach scale across the 5 streams for 1 y. Our 3rd objective was to estimate the contribution of artificial surface-connected macropores to baseflow transient storage.

METHODS

Field surveys of natural macropores

We surveyed surface-connected macropores in 5 streams in the Appalachian Mountains near Blacksburg, Virginia, during base flow in the summer of 2012. The streams varied in channel size, sediment type, and land use (Table 1). For each stream channel, we delineated 2 reach lengths for survey data collection: one reach was 20× the bankfull width in length (short reach) and the 2nd was 60× the bankfull width (long reach). The short reaches were nested at the center of the long reaches.

Table 1. Stream sites surveyed for natural macropores. Watershed land use was calculated by delineating stream watersheds from 1 : 24,000 US Geological Survey (USGS) topographic maps and inspecting aerial photographs. Stream order was calculated using the Strahler method (Strahler 1952) from a 1 : 24,000 USGS topographic map. Ag = agricultural.

Stream name	Land use	Stream order	Short-survey reach length (m)	Long-survey reach length (m)	Bankfull width (m)	Bankfull height (m)	Typical bank sediment type	Reach gradient (m/m)
Slate Branch	Urban	2	104	312	5.2	0.69	Sandy loam	0.0098
Stroubles Creek	Urban/Ag	3	114	342	5.7	0.72	Silt loam	0.0093
Craig Creek	Forest	3	164	492	8.2	0.76	Sand and silt with bedrock and cobbles	0.0092
Poverty Creek	Forest	2	106	318	5.3	0.68	Sandy loam	0.0129
Tom's Creek Tributary	Forest/Ag	2	36	108	1.8	0.43	Silt and clay	0.0023

We measured the spatial dimensions of all macropores in the short reaches. For each macropore encountered, we measured cross-sectional height and width at the opening, length (into bank) to first bend, and height of opening above the thalweg. A dowel rod with metric graduation was used to determine the lengths. Macropore dimensions were measured to the nearest 0.5 cm, such that the greatest error in a single measurement was 0.25 cm. We measured the longitudinal position of each macropore and macropore frequencies in the long reaches, which we used for frequency analysis to characterize macropore abundance along streams.

We surveyed macropores that were ≥ 1 cm in opening height or width. This cutoff was chosen for practical reasons because smaller macropores were difficult to locate. In comparison, in prior studies, macropores have been defined as pore spaces with diameters >0.3 and 0.1 cm according to Beven and Germann (1982) and Luxmoore (1981), respectively. We surveyed macropores on the stream bed and banks, including above and below the baseflow water level.

We measured bankfull width and height at several riffles based on accepted bankfull indicators (Harrelson et al. 1994, Rosgen 1996, McCandless 2003) and averaged along each short reach. We also surveyed channel cross-sectional geometry at every major morphological feature longitudinally along the channel (e.g., break in slope, riffle-to-pool transition) to use as channel geometry in Hydrologic Engineering Centers River Analysis System (HEC-RAS) models of stream hydraulics (see hydroperiod analysis).

Hydroperiod analysis of natural macropores

Macropore hydroperiods refer to the time periods when macropore openings were inundated and, therefore, hy-

draulically connected to the stream. We installed stage gauges midway along each short reach and recorded water levels at 15-min intervals for 1 y with HOBO (Onset, Bourne, Massachusetts) or Levellogger Junior (Solinst, Georgetown, Canada) pressure transducers. Barometric pressure was recorded every 15 min at Stroubles Creek (within a 16-km radius of all sites) and used to correct water levels. We developed rating curves for each stream by calculating a variety of discharges via the velocity–area method using a Marsh–McBirney FlowMate velocimeter (Marsh–McBirney, Frederick, Maryland). We created unsteady flow HEC-RAS stream hydraulics models of each short reach (Brunner 2010a, b) and modeled stream discharge at 15-min intervals for our year of data. We created model channel geometries from short-reach survey data at each site. Bed and bank Manning's n values were back-calculated from discharge measurements using Manning's equation for ≥ 5 storms at each stream reach. We averaged n values from the largest recorded discharge, the smallest recorded discharge, and a mid-range discharge, and used this average for the n of the main channel. Average n was 0.058, 0.072, 0.057, 0.054, and 0.061 for Stroubles Creek, Poverty Creek, Tom's Creek Tributary, Craig Creek, and Slate Branch, respectively. We set overbank n values based on Brunner (2010a) and visual observation to range from 0.08 to 0.11 across sites. The upstream boundary conditions were time-varying discharges over the year, computed from water levels from the stream gauges and the rating curves. The downstream boundary conditions were normal depth using average channel slope. We exported time-varying water-surface elevations at each cross-section from HEC-RAS and used Excel macros to compare these to macropore elevations along the reach. If the modeled water level equaled or exceeded the elevation of the bottom of the macropore opening, we considered the macropore inundated.

Artificial macropore effects on baseflow transient storage

We assessed the importance of macropores for transient storage at base flow by doing 2 tracer-injection experiments, one without macropores (nonmacropore [NM]) and one with artificially created macropores (macropore [M]). We ran a transient-storage experiment on the NM and M scenarios, injecting a conservative tracer at the upstream part of the reach and monitoring the arrival of the tracer downstream (Bencala and Walters 1983, Aumen 1990). The transient-storage experiments were performed on a tributary to Stroubles Creek in Blacksburg, Virginia. The study reach was 35 m long with bed and banks composed of sand, silt, and clay. Small surface storage zones were noted on the outside of 2 meanders with minor eddying of surface water. Stream slope was 0.005 m/m, and baseflow wetted-channel dimensions ranged from 30 to 40 cm wide and 6 to 10 cm deep. We calculated in-stream discharge using dilution gauging (Moore 2004), and results varied little between the 2 experiments and between 5 m and 35 m downstream, averaging 3.3 L/s. We injected a NaCl tracer at the upstream end of the reach and monitored electrical conductivity (specific conductance) 10 m upstream (background), 5 m downstream, and 35 m downstream using an AquaTroll-100 (In-Situ, Fort Collins, Colorado). The first 5 m downstream of the injection was used as the mixing length, so the effective experimental reach length was 30 m. We used a 45-min constant rate drip of 120 mL/min for each experiment, increasing the in-stream specific conductance to 378 $\mu\text{S}/\text{cm}$ (58.3% increase) and 401 $\mu\text{S}/\text{cm}$ (57.7% increase) from background (239 and 254 $\mu\text{S}/\text{cm}$, respectively).

We ran the 1st injection experiment on the NM scenario. Some macropores existed naturally, so we closed these macropores by collapsing them at their connection to the surface stream or plugging them with clay. We ran the 2nd injection experiment after we created artificial macropores. We created these macropores by drilling a 2.54-cm diameter augur bit 30 cm, on average, into the bed and banks in the saturated zone (i.e., below the channel water line). The resultant artificial macropores ranged from 3 to 5 cm in opening diameter. Macropores were drilled such that their orientation was perpendicular to the direction of stream flow or angled slightly upstream (consistent with observation of natural macropores during field surveys). We drilled a total of 100 macropores in the 30-m reach and spaced them approximately evenly along the reach (~ 3 macropores/m). The total macropore volume for the M treatment was 0.0377 m^3 . Several meters of the reach could not retain macropores because of low bank height and sandy soil texture. We clustered macropores upstream or downstream of these areas.

We estimated transient-storage parameters (TSPs) using the One Dimensional Transport with Inflow and Storage

model (OTIS) (Runkel 1998), which uses the 1-dimensional advection–dispersion equation for stream channels with inflowing ground water (gaining from catchment) to simulate exchange with a storage zone. The transient-storage zone can represent either surface-water dead zones or the hyporheic zone. In the OTIS model, A is the channel cross-sectional area (m^2), A_s is the storage-zone cross-sectional area (m^2), D is the dispersion coefficient (m^2/s), and α is the exchange coefficient with the storage zone ($1/\text{s}$) (Runkel 1998). We used the parameterization component of OTIS (OTIS-P) to fit the model to observed data and estimate TSP, α , A , A_s , and D , for the NM and M treatments. To assess whether adding transient-storage parameters A_s and α improved model fit, we also ran OTIS with these parameters turned off. We assumed lateral inflow of ground water was negligible because our experimental reach was short (30 m) and the banks had low-permeability sediments. This assumption was confirmed by negligible differences in discharges from dilution-gauging between 5 and 35 m downstream. Optimization was initially done by allowing all parameters (α , A , A_s , and D) to change. Holding particular parameters constant and re-running the model did not change model results.

Average surface-water channel dimensions from OTIS-P indicated surface-water volume along the reach was 0.651 m^3 , which agreed with cross-sectional surveys. F_{med} is a storage metric used to describe the fraction of median travel time spent in the transient-storage zone. We calculated F_{med} following Runkel (2002)

$$F_{med} = (t_{med} - t_{med}^m) / t_{med} \quad (\text{Eq. 1})$$

where t_{med} is the total median travel time and t_{med}^m is the median travel time in the main channel. We calculated t_{med} and t_{med}^m directly by running models with and without transient storage. The Damköhler number (DaI) compares the average in-stream transport time to the time-scale of transient-storage-zone interaction (Bahr and Rubin 1987, Wagner and Harvey 1997, Harvey and Wagner 2000). We calculated DaI as

$$DaI = \alpha(1 + A/A_s)x/v \quad (\text{Eq. 2})$$

where x is the length (m) of the stream and v is the average stream water velocity (m/s). We also calculated the parameter sensitivity ratio (PSR), which describes the standard deviation of the estimated TSPs (Jin and Ward 2005). We report the 95% confidence interval (CI) limits from OTIS-P for each estimated parameter and the residual sum of squares from the fit of all the parameters together.

RESULTS

A range of types of macropores was observed along stream channels at our 5 survey reaches and other streams (Fig. 1A–D). Crayfish (Fig. 1A) and mink (Fig. 1B) were seen near macropores. At some locations, several probable creation mechanisms were evident, including root decay/erosion, bank erosion, and macroinvertebrate burrows (Fig. 1C). Macropores existed in a range of sediment types, even when gravel was present in the sediment matrix (Fig. 1D). However, a cohesive quality resulting from at least some fine sediment was observed for most matrices surrounding macropores and enabled the macropore to hold its shape.

Spatial characteristics of natural macropores

A total of 474 macropores were characterized in the short reaches, with 112, 111, 106, 107, and 38 macropores found at Slate Branch, Stroubles Creek, Craig Creek, Poverty Creek, and Tom's Creek Tributary, respectively. Macropore openings were slightly wider than tall, with median cross-sectional widths and heights of 3.5 and 3.0 cm, respectively (Figs. 1B, 2A, B). The median macropore length (into bank) to first bend was 15.0 cm (Fig. 2C). Macropores were common above the average bankfull elevation at Craig Creek and Poverty Creek (Fig. 2D) because of the large range of bank heights at these sites (banks were not uniform height), e.g., at Craig Creek one bank

was consistently higher than the other. At other sites, nearly all macropores were below bankfull elevation, with occasional outliers at greater heights (Fig. 2D).

Macropores were unevenly distributed longitudinally along the reaches and were clustered (Fig. 3A–E). Thus, the median interspacing of macropores across all streams (0.38 m) was less than the mean interspacing (1.12 m) (Fig. 2E). Median elevation of macropores above thalweg increased with bankfull width (Fig. 4), indicating that macropores can exist at a large range of bank heights and are not confined to a certain region on the banks. We collected geomorphological data for each macropore survey reach to test for a relationship between macropore distribution and morphological features, but we obtained inconclusive results.

Hydroperiods of natural macropores

We used the variation of macropore opening inundation throughout the year with HEC-RAS hydraulics modeling to calculate the fraction of macropores inundated as a function of discharge (Q) at each site (macropore activation curves, Fig. 5A–E). We also used stage and flow data to calculate the probability of exceeding various discharges (flow duration curves, Fig. 5A–E). We calculated baseflow discharges by averaging nonstorm water levels and using rating curves to estimate a baseflow mean annual Q at each site. One to 32% of macropores were inundated at base

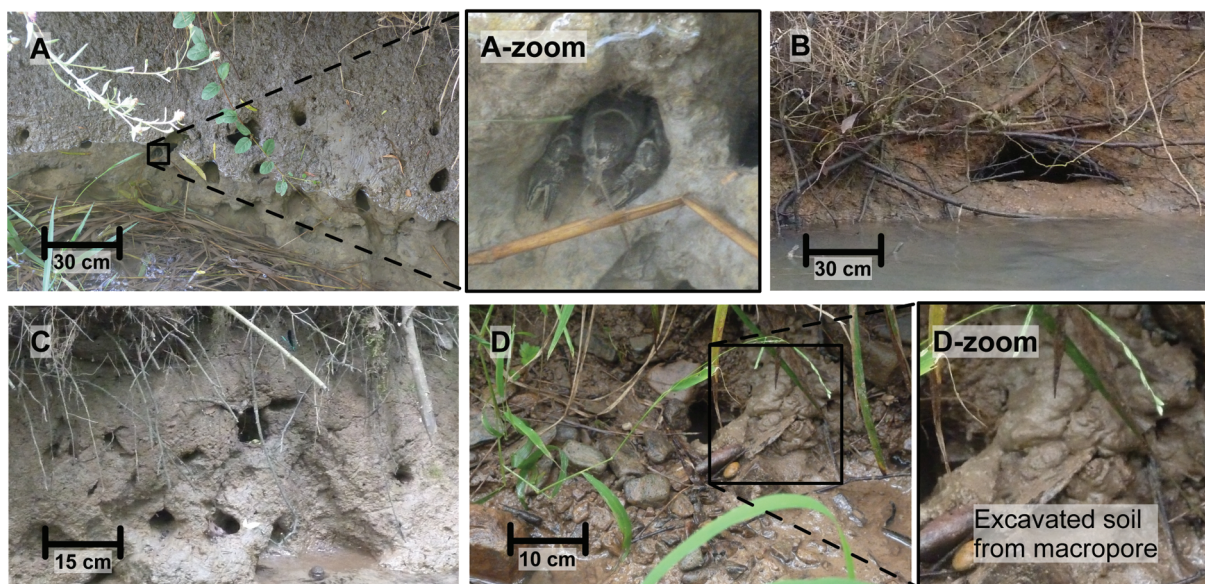


Figure 1. Examples of macropores that probably differ in origin, including macroinvertebrates (crayfish, shown in A-zoom) on the outside of a meander bend (A), vertebrates (e.g., mink) on the banks at a pool (B), root decay/erosion (upper macropores), bank erosion (middle/lower noninundated macropores), and macroinvertebrates (inundated macropores) on banks at a riffle (C), and macroinvertebrates in a sediment matrix containing gravel at a riffle (D-zoom shows recently unearthed soil from the saturated zone probably by crayfish around macropore opening, whereas other nearby soil is more consolidated and drier) (D).

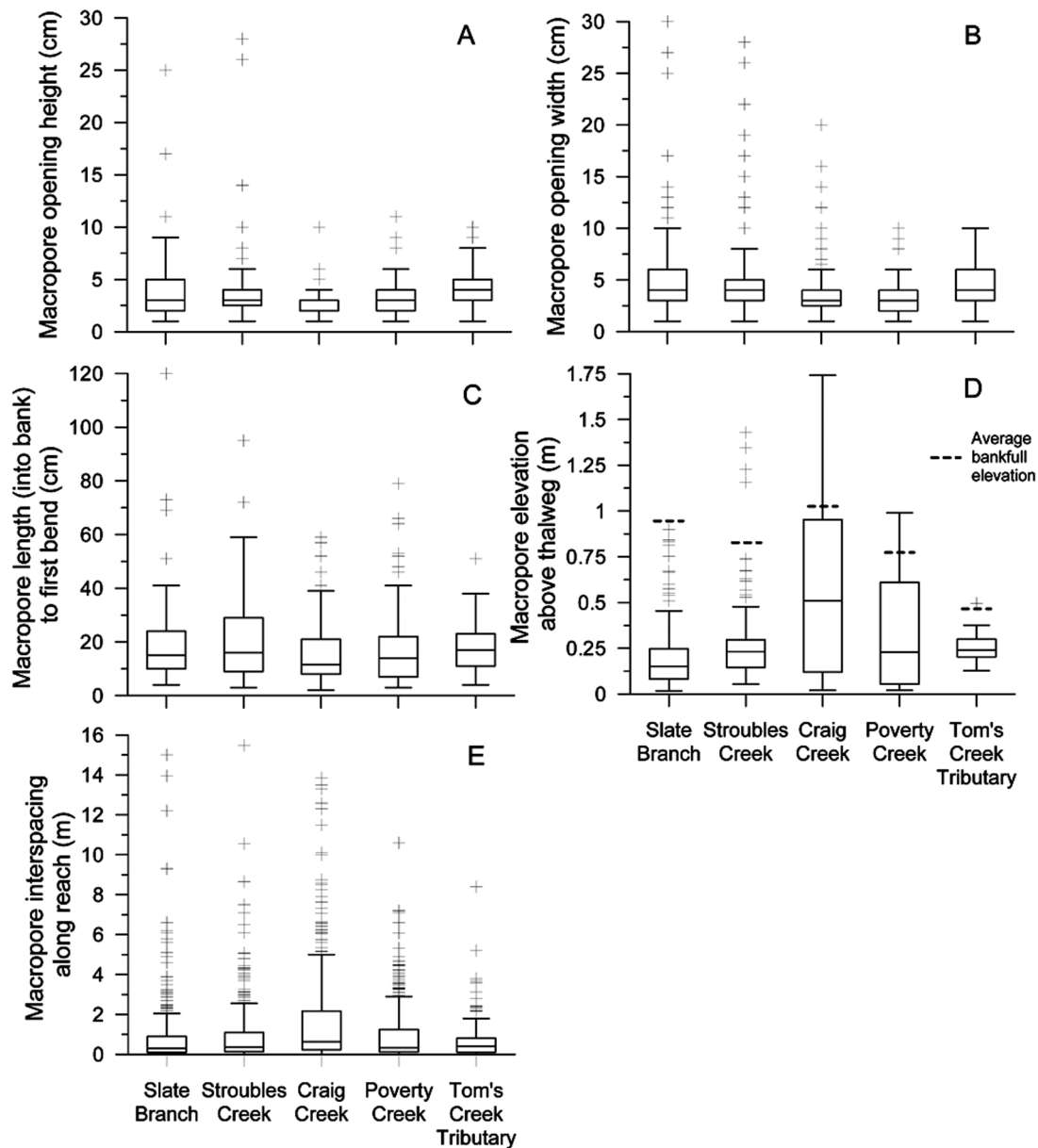


Figure 2. Box plots of macropore opening height (A) and width (B), length into bank to first bend (C), elevation above thalweg (D), and interspacing (E) across field sites. Lines in boxes are medians, box ends are quartiles, whiskers show $1.5\times$ the interquartile range, and crosses are outliers.

flow across sites, but nearly all macropores were inundated during the largest storms. Up to 97% of the macropores across all sites were inundated at some point during the year. Macropore activation plots also identify the probability that a particular percentage of macropores will be inundated. For example, at Stroubles Creek, $Q \geq 0.2 \text{ m}^3/\text{s}$ occurred 18% of the time over the course of the year, and during this time $\geq 36\%$ of the total macropores were inundated (Fig. 5D). Hydroperiod analyses do not consider those macropores that discharge water at base flow as inundated ($<2\%$ of total macropores).

The slope of the macropore activation curve is a function of macropore height on the banks, channel geometry, and stream hydrology. Steep macropore activation curves at low Q values indicate that many macropores were low on the banks or that channel geometry was narrow and deep (i.e., small increases in Q caused a large rise in water level). On the other hand, flatter macropore activation curves indicate that fewer macropores were within a given height on the banks or that channel geometry was wide (i.e., small increases in discharge caused a small rise in water level). Activation curves for Craig Creek leveled off

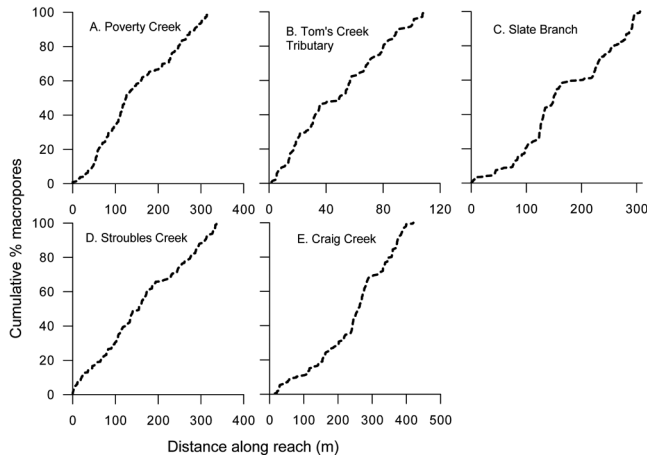


Figure 3. Longitudinal distribution of macropores. Plots show cumulative distribution of macropores along each reach.

at lower values, when ~40% of the total macropores become inundated because the channel has one bank that is lower than the other, so macropores above the shorter bank height require significantly higher discharges to become inundated (Fig. 5E).

Impacts of artificial macropores on baseflow transient storage

Modeled (OTIS) and observed tracer breakthrough curves are shown in Fig. 6. Results show a larger tail for the M than the NM scenario, suggesting an increase in transient storage. Including A_s and α improved the fit of the model to observed data in both treatments, in particular at the rising limb before plateau and the tail of the breakthrough tracer curves. For the NM treatment, the residual sum of squares (RSS) was 27.3 without tran-

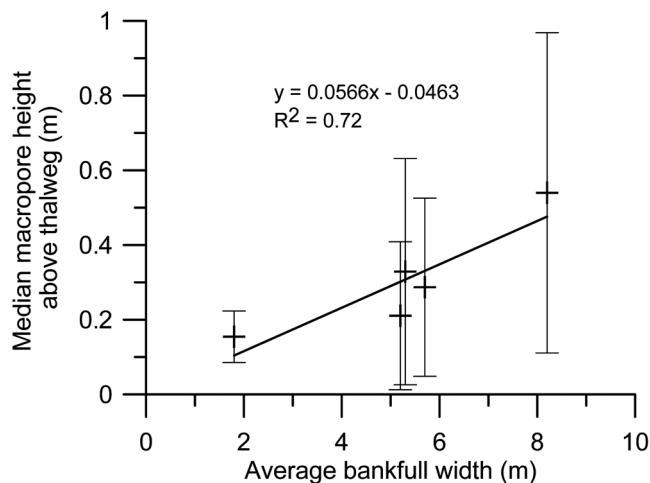


Figure 4. Relationship between median (± 1 SD) height of macropore above thalweg and average channel bankfull width. Only median values were used in the relationship.

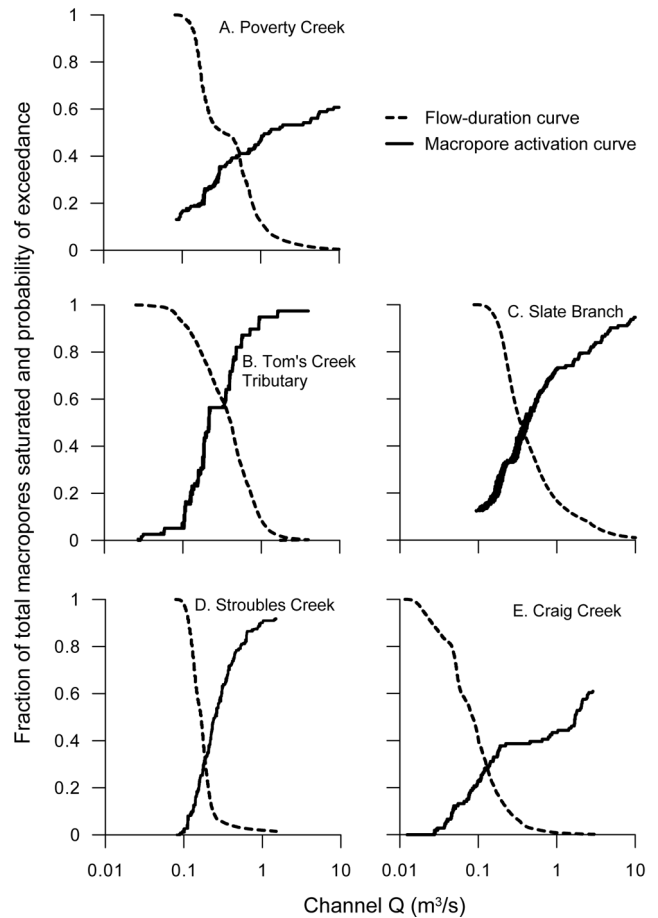


Figure 5. Hydroperiod analyses of natural macropores in Poverty Creek (A), Tom's Creek Tributary (B), Slate Branch (C), Stroubles Creek (D), and Craig Creek (E). Maximum discharges shown are $<2\times$ the maximum discharge measured with the velocity-area method to avoid extrapolating to higher discharges from rating curves and to show trends in macropore activation curves at lower discharges in more detail. Macropore activation curves are based on macropore inundation over time.

sient storage and 0.216 with transient storage. For the M treatment, the RSS was 43.9 without transient storage and 0.327 with transient storage.

A_s , A_s/A , α , and F_{med} were all higher (16.6, 15.0, 20.9, and 31.5%, respectively) with macropores present (M) than without (NM) (Table 2). Overall, these results indicate a modest but measurable effect of the artificially created macropores. In contrast, D and A varied $<3\%$ between the 2 treatments, consistent with minimal changes to the channel geometry and Q . Q decreased slightly between treatments, from 3.4 L/s to 3.2 L/s.

We used several statistical measures to quantify the uncertainty of our results. The Dal for the NM and M treatments were 3.99 and 4.54, respectively. $Dal = 1$ is ideal to ensure reliability of A_s and α (Wagner and Harvey 1997, Harvey and Wagner 2000), but is difficult to

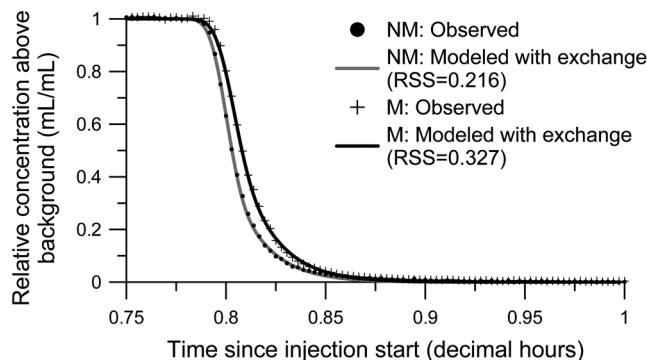


Figure 6. Falling-limb breakthrough tracer curves (BTCs) shown in relative concentration, which is defined as volume of injectate divided by volume of stream ($\times 1000$). Relative concentration is also normalized to the plateau concentration of each injection. Observed BTCs and BTCs modeled (OTIS) with exchange 30 m downstream of injection are shown for no macropore (NM) and macropore (M) treatments. RSS = residual sum of squares.

achieve given that it is highly dependent on reach length and channel discharge. A broader range of $0.1 < DaI < 10$ is commonly accepted (Jin and Ward 2005, Wondzell 2006, Becker et al. 2013), indicating that our A_s and α estimates are reliable. Another commonly used metric of certainty is PSR. In our analyses the PSR was >2 for all cases, suggesting good certainty in parameter estimates (Jin and Ward 2005).

We used CIs to quantify the statistical significance of the effect of artificial macropores. The 95% CIs for the 2 treatments overlapped substantially for α , such that the CI for the M treatment included the mean optimized α value for the NM. This large overlap indicates that the difference in α was not significant at the 95% confidence level. Neither mean optimized value of A_s fell within the CI of the opposite treatment, but the CIs overlapped slightly. We concluded that macropores did not significantly increase A_s at the 95% confidence level. Nevertheless, our results suggest that macropores increased transient storage in a reach that otherwise had little potential for hyporheic exchange (see Discussion).

DISCUSSION

Natural macropore distribution and abundance

Surface-connected macropores are widespread and common, at least within the Valley and Ridge physiographic province of the Appalachian Mountains in southwestern Virginia (Figs. 1A–D, 2A–E). We found macropores throughout the long reach of each of the 5 streams, despite variation in sediment texture, land use, stream size, and gradient. Thus, the presence of surface-connected macropores appears to be typical rather than anomalous. The presence of at least some fines within the sediment matrix of all 5 streams provides a cohesive quality to the

sediment matrix that appears to help hold the shape of the macropore. The prevalence of macropores in our study suggests they are an important dimension of aquatic habitat, and may be common elsewhere.

Hydrologic implications of macropore characteristics

The hydrologic importance of individual macropores will depend upon their dimensions (e.g., size, length) and connectivity with other macropores or preferential flow paths. Such connectivity has been studied in groundwater fractures where the better connected the system of fractures, the higher the flow rate through the system (Matthai and Belayneh 2004). Similarly, Noguchi et al. (1999) showed that short macropores in forested hill slopes often were connected to larger systems of preferential flow paths and could strongly influence hillslope hydrology. Connectivity can depend on many geometric variables, such as macropore length, orientation (angle) in the banks, and interspacing. In particular, macropores with low interspacing, longer lengths, and different orientations may be more likely to be connected.

Ours appears to be the first systematic study of surface-connected macropore dimensions in multiple streams. Jones and Cottrell (2007) surveyed macropores (soil pipes) along a stream in England over time and found macropore interspacings from 4.75 to 15.75 m, mean diameters from 5.0 to 10.5 cm, and mean heights above thalweg from 24.2 to 56.6 cm. Our spatial dimensions and interspacing are smaller than theirs, perhaps because they focused on soil pipes, which are one of the larger types of macropores. We note that even our greater observed macropore densities probably are underestimates because we did not include macropores with an opening diameter <1 cm.

We measured macropore length (into bank) to first bend because it was an efficient method of characterizing large numbers of macropores with minimal damage to their natural state. This approach is a useful starting point but underestimates macropore tortuous length, and probably underestimates macropore hydrologic significance, including connectivity with other preferential flow paths. Macropores can be quite tortuous, e.g., macropores created by crayfish or mitten crabs (Williams et al. 1974, Helfrich et al. 2009, USGS 2012). In particular, mitten crab burrows along the Sacramento River have been cast in resin and shown to extend horizontally 87 cm before dipping down at a 30° angle and extending another 105 cm (USGS 2012).

We suggest that future investigators should explore innovative methods to detect macropores and to quantify their spatial dimensions, particularly tortuous length and connectivity more accurately. For example, electrical resistivity imaging has been used successfully to identify macropores in a meander bend (Menichino et al. 2014), ground-penetrating radar was used in the riparian zone to detect macropores (Holden 2005, Gormally et al. 2011),

Table 2. Stream and transient-storage parameters from the artificial macropore experiment. Mean velocity (ν) was calculated from arrival of 50% of plateau concentration. σ = standard deviation, Q = discharge, D = dispersion coefficient, A = channel cross-sectional area, A_s = storage-zone cross-sectional area, α = exchange coefficient with the storage zone, Dal = Damköhler number (see text for calculation), F_{med} = fraction of median travel time spent in the storage zone, BTC = breakthrough tracer curve, PSR = parameter sensitivity ratio, TSP = transient-storage parameter.

Statistic	Q(L/s) from dilution gauging	ν (m/s) from BTC	Transient-storage parameters and analysis						
			D (m ² /s)	A (m ²)	A_s (m ²)	α (1/s)	Reach Dal	F_{med}	A_s/A
No macropores (NM)	3.38	0.182	4.45×10^{-2}	1.85×10^{-2}	2.83×10^{-3}	3.21×10^{-3}	3.99	0.0588	0.153
PSR : TSP/ σ			14.8	157.1	27.5	11.7			
95% lower confidence limit			3.86×10^{-2}	1.83×10^{-2}	2.62×10^{-3}	2.67×10^{-3}			
95% upper confidence limit			5.05×10^{-2}	1.88×10^{-2}	3.03×10^{-3}	3.75×10^{-2}			
Macropores (M)	3.20	0.171	4.58×10^{-2}	1.87×10^{-2}	3.30×10^{-3}	3.88×10^{-3}	4.54	0.0773	0.176
PSR : TSP/ σ			10.9	97.8	20.0	8.8			
95% lower confidence limit			3.75×10^{-2}	1.83×10^{-2}	2.98×10^{-3}	3.00×10^{-3}			
95% upper confidence limit			5.41×10^{-2}	1.90×10^{-2}	3.63×10^{-3}	4.75×10^{-3}			
% change	-5.6	-6.0	2.9	1.1	16.6	20.9		31.5	15.0

and liquid latex has been used to map diverse networks of macropores in the unsaturated zone (Abou Najm et al. 2010).

Hydrologic contributions of macropores to stream discharge at base flow

Macropores act as preferential flow paths and, therefore, may increase rates of transport from the floodplain/riparian zone to streams during base flow. For example, Bohlke et al. (2007) showed that macropores could provide the majority of base flow to their study stream. We expect that seasonal changes in water-table elevation will change the number of macropores that provide water during base flow, and the direction of transport will depend on applied head gradients. For example, a macropore discharged into Slate Branch during base flow in winter (Fig. 7A), but not during base flow in late summer 2012.

Macropores discharging to the stream may affect stream water quality. For example, macropore discharge that is substantially warmer or cooler than channel discharge can affect channel water temperature. Macropores can discharge water that is 1.7°C colder (Fig. 7B) and 1.6°C warmer (Fig. 7D) than channel discharge. Whether base flow warmed or cooled the surface stream depended on the daily stream temperature, which fluctuated with ambient air temperature. Individual macropores may not

cause a substantial change in bulk downstream water temperatures, but local effects may be significant and could influence rates of local biogeochemical reactions or provide unique habitat for aquatic organisms. Discharging macropores also may be significant for biogeochemistry. For example, macropores in hill slopes or floodplains can rapidly transport runoff that is high in nutrients to the stream, bypassing denitrification hotspots in the sediment matrix (Bohlke et al. 2007). In the midwestern USA, tile drainage can transport pollutants to streams and control baseflow discharge (Hallberg 1987, Schilling et al. 2012). In such situations, macropores are actively bypassing the attenuative capacity of the hyporheic zone. Whether discharging macropores have a beneficial or adverse impact on water quality and stream functions will depend on hydrologic conditions, spatial characteristics of macropores, and physiochemical properties of surface water and ground water.

Effects of macropores on baseflow transient storage

We studied a reach in a small stream with bed and banks composed of sand, silt, and clay, and an average travel time of 3 to 4 min. As a result, preexisting hyporheic exchange and transient storage were limited. A_s for the NM treatment was 0.00283 m², which is at the low end of a naturally observed range of 0.001 to 1 m² from 136 different studies (Stofleth et al. 2008). We attribute the low A_s in the NM treatment to a combination of low-permeability bed and banks, little geomorphic complexity to create hyporheic gradients, low surface discharge (on average 3.3 L/s), and small cross-sectional area of flow. However, NM α (0.00321/s) was within the upper ½ of the range of natural observed values (10⁻⁵–10⁻² L/s) (Stofleth et al. 2008), which we attribute to exchange with existing surface storage zones. Exchange rates with surface transient-storage zones can be more than an order of magnitude higher than exchange rates with hyporheic transient-storage zones (Briggs et al. 2009, 2010). We attribute such surface transient storage to exchange with small surface storage zones on the outside of meanders with eddying of water (Briggs et al. 2010). Average D (0.0452 m²/s) was within an accepted range (Jin and Ward 2005, Wondzell 2006, Stofleth et al. 2008). The average A (0.0186 m²) agreed with our measured channel dimensions. Interactions between transient-storage parameters could cloud process-based meaning, but we think each parameter is more indicative of certain real hydrologic processes than others and these differences can help relate these parameters to processes in our stream.

Our results suggest a trend of increased transient storage with constructed macropores. The increase in F_{med} suggests that the artificial macropores increased the fraction of stream travel time spent in the transient-storage zone (Table 2). Our results did not indicate significant increases in A_s or α at the 95% confidence level, but both parameters

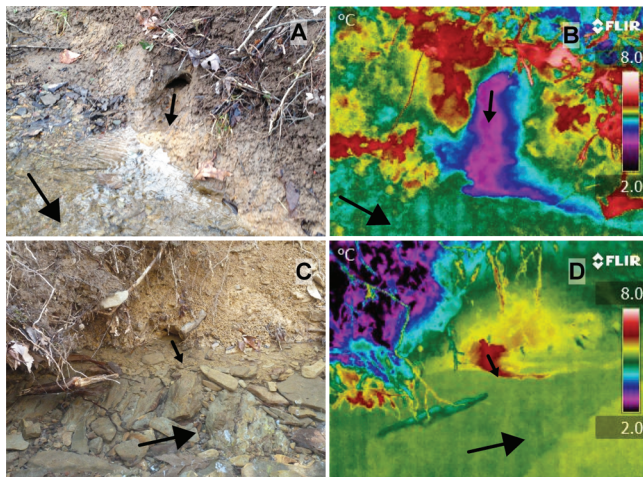


Figure 7. Thermal discharges from macropores during winter. A.—Macropore opening that is 4 cm tall $\times$ 3.5 cm wide discharging water into Slate Branch. B.—Forward Looking Infrared (FLIR) image of the same macropore, discharging water that is 1.7°C colder than stream flow on 23 February 2013 (air temperature = 4.4°C). C.—Macropore opening that is 3 cm tall $\times$ 5 cm wide at Poverty Creek with no observable flow. D.—FLIR image of same macropore actively discharging water 1.6°C warmer than stream flow on 25 February 2013 (air temperature = -0.56°C). Surface water and groundwater flow directions shown by larger and smaller arrows, respectively. The average air temperature for the month of February was 0.72°C.

did increase in a system that otherwise had limited potential for hyporheic exchange. This increase may have been caused by macropores serving as surface storage zones or as unobstructed pathways for transport to the subsurface. However, A_s was low even after macropore creation, a result suggesting that the net effect of macropores on hyporheic exchange was limited by low-permeability silts and clays lining the macropores. We note that the drill bit used to construct macropores could have smeared clay in the artificial macropore, thereby sealing it off from existing, natural preferential flow paths. Future investigators should test whether macropores have a larger effect on transient storage where sediments are coarser or macropores extend further into the banks, particularly where macropores penetrate lower-permeability surface layers to reach other preferential flow paths (e.g., paleochannel, other macropores).

We increased the surface volume of this small stream by 6.8% by creating 100 macropores along 30 m. This means that we added a macropore, on average, every 0.30 m. Comparatively, our survey data across 5 streams indicate an average natural macropore interspacing of 1.12 m. Therefore, we created 3.7× more artificial macropores than we observed for natural macropores in the field. Nevertheless, macropores are often clustered (Fig. 3A–E), and we saw a median interspacing of macropores of 0.38 m in our survey of 5 streams. This value is close to our construction interspacing of 0.30 m. Thus, clusters of macropores may cause effects on transient storage in natural systems similar to what we observed, but over short reaches. We acknowledge that only 1 to 32% of natural macropores were active at base flow. Therefore, in natural systems, macropores may have less effect on baseflow transient storage than we observed.

Macropore hydrologic significance across temporal scales and methodological limitations

From our hydroperiod analysis we found that 1 to 32% of macropores were inundated at baseflow, but that >97% of macropores were inundated at all sites during the largest of storms in a 1-y period. Therefore, channel water can access most surface-connected macropores only during storms. Thus, such macropores have potential to enhance bank storage when rapid reversals of head gradients at stream margins push water into and out of the banks (Sawyer et al. 2009). This concept also applies to water-level fluctuations caused by daily snowmelt, evapotranspiration, hydropeaking, and tidal fluctuations (Peterson and Connelly 2001, Westbrook et al. 2005, Arntzen et al. 2006, Lohede and Lundquist 2009, Sawyer et al. 2009, Francis et al. 2010, Wondzell et al. 2010). Bank storage can have significant effects on temperature and biogeochemistry of the stream and the floodplain ground water (Chen and Chen 2003, McCallum et al. 2010, Ander-

son et al. 2011, Gu et al. 2012). As stream levels rise during storms, hydraulic gradients between channel and riparian groundwater that drive bank storage can increase beyond their levels at base flow, but Q in the channel may increase faster than transport through macropores caused by newly inundated macropores. The net effect of these processes is unclear and is an important area for future research.

Our hydroperiod analysis had some methodological limitations that may influence the spatial and temporal accuracy of the macropore activation curves (Fig. 5A–E). First, we presented the macropore-activation curves for 1 y to provide a general understanding of how macropore inundation can vary over discharge and among streams. If data were collected for more years, annual changes in stream hydrology and variation in flow-duration curves would influence which macropores were saturated. Storm flows are not evenly distributed over a year, so macropore importance also may vary seasonally. For example, the greatest number of storms in the streams we studied occurred in winter, and therefore, macropore activation was highest during winter. Second, we defined macropore saturation based on surface-water elevation in our macropore hydroperiod analysis. Therefore, the hydroperiod analysis is conservative because it does not account for the small minority (<2%) of macropores that we observed discharging water during non-storm conditions. The percentage of discharging macropores probably varies in time.

A 3rd limitation is that our HEC-RAS models of stream hydraulics did not include obstructions or in-stream structures in the channel that can influence water levels. As a result, macropores upstream of flow obstructions may be activated at lower discharges than we predicted. Fourth, we calculated Manning's n from observed discharges for several storm events and applied this value to the entire main channel. However, Manning's n can change with local conditions, such as vegetation and sediment texture, in the channel. For example, sections of the channel with higher Manning's n will experience higher water levels and higher levels of macropore activation. Fifth, we identified only macropores that were ≥ 1 cm in diameter. Including macropores smaller than those studied here may shift the trends we observed. These methodological limitations and considerations could increase the complexity of the macropore-activation curve. Nevertheless, our work demonstrates the concepts and presents useful preliminary data on how macropore inundation varies with flow and among streams.

ACKNOWLEDGEMENTS

We thank the Thomas F. Jeffress and Kate Miller Jeffress Memorial Trust and the Institute for Critical Technology and Applied Science at Virginia Tech for support to Garrett Menichino. We also acknowledge the support of the National Science Foundation (NSF) through NSF Research Experience for Undergraduates Site Grant EEC-1062860. Any opinions, findings, and conclusions or

recommendations expressed in this paper are those of the authors and do not necessarily reflect the views of the NSF. We thank Kate Aulenbach for field assistance.

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